HACKATHON TECHNICAL DESIGN

Decentralized IoT Edge Swarm Task Allocation and Self-Healing Mesh Simulator

A mathematical architecture for CubeSat constellations and aerial IoT networks

Core synthesis
Resource-aware distributed optimization + Shen-style local schedule repair + contact-aware routing + bounded glowworm search + explicit task ownership + independent safety validation.

Complete design specification

State and sensor models; coupled constraints; optimization updates; fault recovery; ownership semantics; pseudocode; computational budgets; worked example; and evaluation plan.

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Status: proposed research and simulator architecture. No implementation, flight qualification, global optimality proof, or new experimental performance result is claimed.

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1 Purpose and design principle

Next-generation autonomous sensor constellations must coordinate observation and delivery without depending on continuous cloud or ground-station control. Independent greedy sensing can duplicate captures, exhaust batteries, fill flash buffers, and overload relay nodes. This specification compiles the preceding design into a consistent mathematical architecture grounded in the supplied papers and assignment [1, 2, 3, 4, 5].

Recommended architecture: state estimation and fault detection → robust eligibility screening → distributed allocation → bounded local schedule repair → contact-aware routing → ownership and reservation → execution-time safety validation.

The reference optimization is deliberately expressive; the online implementation solves only bounded, affected fragments. Robustness comes from uncertainty margins, resource accounting, explicit failure semantics, and safe fallbacks. Complexity by itself supplies no robustness guarantee.

1.1 Source contribution and limitations

Source	Retain	Extend or qualify
Shen et al. [1]	Incremental dependencies; support tasks; locked prefix; adjustable suffix; discrete adaptive PSO.	Add delayed telemetry, node failures, spacecraft physics, resource queues and decentralized ownership.
Glowworm mesh paper [2]	Local adaptive route exploration and load distribution.	Search feasible communication paths, not arbitrary spacecraft positions. Benchmark against deterministic routing.
Monograph [3]	Constraint coupling, consensus ADMM, dual decomposition and constraint exchange.	Apply convergence results only to the subproblem satisfying their hypotheses.
Assignment [4]	Battery-aware assignment and replacement logic.	Verify algorithms independently; replace simplified motion and energy models.
DMAS [5]	Local anomaly evidence and peer assessment.	Separate scoring from Byzantine agreement; use realistic space-link delays.

Shen et al. assume arrivals are immediately observed by the planner and exclude UAV failure during the mission. Their local-repair feasibility arguments therefore do not automatically hold after a failed relay, battery fault, or stale measurement.

The assignment's numerical summary reports objective values 8504.7409 (Hungarian), 23915.2735 (MUR), 21338.6198 (MURD), and 25854.8241 (MURID), despite text claiming matching optima. Its supplied routines use greedy seeding and swap refinements rather than the stated ADMM iterations; robot positions are directly updated to assigned stations. Its formulation is useful, but these runtime and optimality claims are not adopted here.

Reputation-weighted voting is not itself Byzantine agreement. A replicated authorization ledger needs an actual authenticated agreement protocol, including its membership, ordering, quorum and recovery rules [7]. Likewise, the mesh paper's reported performance is not treated as a performance forecast for this architecture.

2 System model and notation

Let $\mathcal{V} = \{1, \dots, N\}$ denote agents. At decision epoch t, the fine planning horizon contains slots $h = 0, \dots, H-1$ of duration Δt. The communication graph and task set vary with time.

Symbol	Meaning
$\mathcal{T}, \mathcal{C}$	Active tasks and nodes in the current repair patch.
r_i, v_i, q_i, ω_i	Position, velocity, quaternion and angular velocity.
E_i, T_i, B_i	Usable battery energy, component temperatures and persistent storage occupancy.
Q_i^{cpu}	Processing backlog, measured in cycles.
$\Pi_i, \Pi_i^{\text{locked}}, \Pi_i^{\text{free}}$	Node plan, committed prefix and adjustable suffix.
x_{ijhm}	Binary observation start: agent i, task j, slot h, mode m.
z_{ifh}	Support-task decision for support function f.
$y_{ik,h}, f_{ik,j,h}$	Communication activation and bytes routed for task j.
d_j, s_j	Successful-delivery indicator and unmet observation requirement.
$\Sigma_i, \mathcal{U}_t$	Estimate covariance and bounded uncertainty set.
$D_{\mathcal{C}}, K_A$	Relaxed patch dimension and allocation iteration cap.
P, K_P, F	PSO population, iteration cap and adjustable schedule length.

2.1 Two physical models, one coordination layer

Use common task, ownership, networking and data-management interfaces, but distinct platform feasibility sets:

$$\mathcal{F}_i = \begin{cases} \mathcal{F}_i^{\text{space}}, & \text{CubeSat}, \\ \mathcal{F}_i^{\text{air}}, & \text{UAV}. \end{cases} \tag{1}$$

A CubeSat selects visibility opportunities, attitude, payload modes, processing, contacts and charging-compatible actions. A UAV additionally selects feasible flight segments, headings, speeds and return-to-base actions. Euclidean routing to a target is not an orbital observation model.

Use a fine horizon for imminent activity and a coarse horizon for eclipse energy, charging recovery and future buffer drainage. Couple the two through terminal resource constraints. A short-horizon planner must not spend energy required during a later eclipse.

3 Sensor interface and state estimation

The local estimated state is

$$s_i = [r_i, v_i, q_i, \omega_i, E_i, \text{SOH}_i, T_i, B_i, Q_i^{\text{cpu}}, h_i^{\text{hw}}, \ell_i^{\text{net}}, \mathcal{P}_i, \mathcal{R}_i]. \tag{2}$$

Here h_i^{hw} describes health/capability, ℓ_i^{net} describes communication state, and $\mathcal{P}_i, \mathcal{R}_i$ are the plan and resource reservations. Every estimate carries $(\hat{s}_i, \Sigma_i, t_i^{\text{stamp}}, \text{validity flags})$.

Group	Inputs	Planning products
Navigation	Position, altitude, velocity, heading, acceleration, GNSS or ephemeris residuals.	Predicted trajectory, visibility, contacts and uncertainty.
Attitude	IMU, star tracker, magnetometer, pointing/tracking error, angular rate.	Slew time, pointing feasibility and image-smear risk.
Power	Voltage, current, SOC, SOH, panel current, irradiance, eclipse status.	Usable energy, conservative charging and reserve.

Group	Inputs	Planning products
Thermal	Battery, CPU, payload and radio temperatures.	Thermal headroom and component derating.
Storage	Raw/output occupancy, scratch demand, flash errors, free blocks, write rate.	Peak buffer demand, retention and write budgets.
Computation	CPU frequency/utilization, queue length, measured execution time, watchdog.	Available cycles and processing deadline risk.
Radio	RSSI, SNR, loss, RTT, Doppler, antenna state, channel occupancy.	Payload rate, contact volume, reliability and energy.
Packet metadata	Source/destination, size, acquisition time, expiry, sequence, priority, content hash.	Delivery obligations, duplicate suppression and data age.
Security	Replay counts, invalid authentication, bursts, unexpected flows.	Anomaly evidence and interface restrictions.
Payload	Ambient light, cloud/occlusion estimate, target confidence, crop/target quality, requested resolution.	Expected quality and mission utility.
Mechanical	Wheel speed/current/vibration, actuator faults, payload deployment; UAV motor/propeller health.	Available pointing, motion and payload modes.
Mission	Authorized alerts, hazards, coverage segment ID, priority and deadlines.	Task generation, preemption and constraints.

Do not insert every raw sample into the optimizer. Use calibrated state estimators and compact derived features. Altitude is derived from position; velocity and linear velocity are not independent states. Missing sensors increase uncertainty or reduce capability rather than producing invented values.

$$s_{i,k+1} = f_i(s_{i,k}, u_{i,k}) + w_{i,k}, \tag{3}$$

$$z_{i,k} = h_i(s_{i,k}) + \nu_{i,k}. \tag{4}$$

Separate navigation/attitude, battery, thermal and link estimators are generally preferable to a single enormous filter. Preserve correlations where they affect joint uncertainty.

4 Task and dependency representation

Represent an observation-and-delivery task as

$$\tau_j = (\mathrm{id}_j, \mathcal{G}_j, [a_j, b_j], d_j^{\mathrm{obs}}, d_j^{\mathrm{deliver}}, V_j, \underline{Q}_j, D_j^{\mathrm{raw}}, \mathcal{M}_j, \mathcal{D}_j, \kappa_j). \tag{5}$$

The tuple specifies target region, window, duration, delivery deadline, value, minimum quality, raw volume, modes, dependencies and permitted independent observation count. Normally $\kappa_j = 1$; stereo or deliberate redundancy can use $\kappa_j > 1$.

Define canonical observation-instance IDs using region, time bin, sensor class and mission version. Different IDs for the same physical observation would defeat duplicate avoidance. Repeated surveillance creates new instances rather than permanently marking a region complete.

Each mode encodes

$$m \mapsto (D_{jm}^{\mathrm{out}}, C_{jm}^{\mathrm{cpu}}, E_{ijm}^{\mathrm{acq}}, Q_{ijm}, d_{jm}^{\mathrm{proc}}, B_{jm}^{\mathrm{peak}}). \tag{6}$$

Compression therefore changes output bytes, CPU cycles, energy, delay and quality together.

4.1 Typed dependencies

Maintain $\mathcal{G}_D = (\mathcal{T}, \mathcal{E}_D)$ with observation-to-processing, processing-to-delivery, relay-to-delivery, downlink-to-future-storage, calibration-to-quality and charging-attitude-to-energy edges.

Support tasks include relay, downlink, compression, calibration, health recovery and charging-compatible attitude. Their utility comes from enabled mission outcomes, not an arbitrary reward for performing empty support activity.

For soft support effects, one possible saturating model is

$$V_j(S) = \bar{V}_j \left[1 - \exp\left(-\beta_j \left[q_j^0 + \sum_{f \in S} w_{fj}\phi_{fj}\right]\right)\right]. \tag{7}$$

The coefficient ϕ_{fj} captures temporal validity. Hard prerequisites instead require $a_j^{\text{accept}} \leq z_f$. Calibration required for a valid image is a prerequisite, not just a utility bonus. A universal $1 - 1/e$ approximation ratio is not asserted: heterogeneous costs, prerequisites, routing and timing alter the needed assumptions.

5 Physical and resource dynamics

5.1 Orbital motion, pointing and UAV motion

For the space model,

$$\dot{r}_i = v_i, \tag{8}$$

$$\dot{v}_i = -\mu r_i / \|r_i\|^3 + a_{J_2,i} + a_{\text{drag,i}} + a_{\text{other,i}}. \tag{9}$$

The reference simulator can use higher fidelity than the onboard predictor, provided omitted perturbations are covered by uncertainty margins. Define

$$\chi_{ijh}^{\text{obs}} = \chi^{\text{LOS}}\chi^{\text{FOV}}\chi^{\text{point}}\chi^{\text{illum}}\chi^{\text{quality}}\chi^{\text{health}}, \qquad x_{ijhm} \leq \chi_{ijh}^{\text{obs}}. \tag{10}$$

A conservative initial slew estimate is

$$t_{i,j\to k}^{\text{slew}} \geq \Delta\theta_{jk}/\omega_i^{\max} + t_i^{\text{settle}}, \tag{11}$$

supplemented by torque, wheel momentum, acceleration and tracking constraints. If acceleration is limited, the constant-rate formula alone is insufficient.

For a UAV use $\dot{p}_i = v_i$, $\dot{v}_i = u_i + w_i^{\text{wind}}$, plus flight envelope, geofence, obstacle, payload capacity, range and feasible landing/return constraints. Heading is a model state or derived quantity according to the selected vehicle dynamics.

5.2 Energy and charging

$$E_{i,h+1} = E_{i,h} + \eta_i^{\text{ch}} P_{i,h}^{\text{solar}} \Delta t - P_{i,h}^{\text{load}} \Delta t - E_{i,h}^{\text{curtail}}. \tag{12}$$

Curtailment is nonnegative unused harvest, bounded by available harvest; it cannot create or remove arbitrary energy. Consistent conversion losses must be included if load is specified at the bus rather than battery terminals.

$$P^{\text{load}} = P^{\text{bus}} + P^{\text{ADCS}} + P^{\text{payload}} + P^{\text{CPU}} + P^{\text{TX}} + P^{\text{RX}} + P^{\text{heater}}, \tag{13}$$

$$P_{i,h}^{\text{solar}} = \chi_{i,h}^{\text{sun}} \eta_i^{\text{panel}}(T) A_i S_h \max(0, n_{i,h}^{\top} s_h), \tag{14}$$

$$E_i^{\text{reserve}}(h) \leq E_{i,h} \leq E_i^{\text{usable,max}}. \tag{15}$$

An imaging attitude can reduce charging and antenna access. The reserve covers essential bus operation, safe attitude, recovery and uncertainty through the relevant eclipse/recovery period. A UAV reserve includes feasible return/landing energy. Never clamp an infeasible predicted battery value and then label the plan safe.

5.3 Thermal behavior

A lumped spacecraft thermal balance is

$$\begin{aligned} C_i^{\mathrm{th}}\dot{T}_i = P_i^{\mathrm{diss}} + Q_i^{\mathrm{solar}} + Q_i^{\mathrm{albedo}} + Q_i^{\mathrm{EarthIR}} + Q_i^{\mathrm{cond}} \\ - \epsilon_i \sigma A_i (T_i^4 - T_{\mathrm{space}}^4). \end{aligned} \tag{16}$$

Radiative temperatures are in kelvin. The external spacecraft model has no atmospheric convection; the UAV model may require it. A reduced onboard predictor is

$$T_{i,h+1} = A_i^T T_{i,h} + B_i^T P_{i,h}^{\mathrm{diss}} + d_{i,h}, \qquad T_i^{\min} \le T_{i,h} \le T_i^{\max}. \tag{17}$$

The superscript T here labels thermal matrices, not a transpose. Use temperature-dependent $f_i^{\max}(T)$ and $P_i^{\mathrm{TX,max}}(T)$.

5.4 Computation

With backlog measured in cycles,

$$Q_{i,h+1}^{\mathrm{cpu}} = \left[Q_{i,h}^{\mathrm{cpu}} + A_{i,h}^{\mathrm{cycles}} - f_{i,h}^{\mathrm{alloc}} \Delta t \right]_+ . \tag{18}$$

Reserve computation for essential control:

$$f^{\mathrm{payload}} + f^{\mathrm{optimizer}} + f^{\mathrm{network}} \le f^{\mathrm{available}} - f^{\mathrm{control,reserve}}. \tag{19}$$

A calibrated dynamic-energy approximation for C executed cycles is $E^{\mathrm{CPU}} \simeq \kappa C f^2$. Processing also has RAM and scratch-storage requirements; persistent flash capacity alone is insufficient.

5.5 Storage, transformations and retention

$$\begin{aligned} B_{i,h+1} = B_{i,h} + D_{i,h}^{\mathrm{acquired}} + D_{i,h}^{\mathrm{received}} + D_{i,h}^{\mathrm{transform,out}} \\ - D_{i,h}^{\mathrm{transform,in,freed}} - D_{i,h}^{\mathrm{ack,freed}} - D_{i,h}^{\mathrm{expired}}. \end{aligned} \tag{20}$$

Require $B_{i,h} \le B_i^{\max} - B_i^{\mathrm{reserve}}$. During processing, validate actual simultaneous buffer lifetimes:

$$B_i^{\mathrm{existing}} + B_i^{\mathrm{raw}} + B_i^{\mathrm{scratch}} + B_i^{\mathrm{output}} \le B_i^{\mathrm{safe,max}}. \tag{21}$$

If buffers are streamed or reused, model that explicitly rather than imposing a fictitious simultaneous requirement. Sender bytes are freed only under the chosen acknowledgement/retention policy. Reserve receiver storage before forwarding; prioritize retention by mission obligation. Bound logs, ring buffers and metadata, and account for flash write limits and detected errors.

6 Time-expanded contact network

Construct $\mathcal{G}_C^H = (\mathcal{V}^H, \mathcal{E}_C^H)$ with node-time vertices, feasible transmission edges and storage/wait edges. The time expansion enforces causal forwarding. Space DTN and contact graph routing are relevant foundations [8].

$$\chi_{ik,h}^{\mathrm{link}} = \chi^{\mathrm{LOS}} \chi^{\mathrm{antenna}} \chi^{\mathrm{linkbudget}} \chi^{\mathrm{radiohealth}} \chi^{\mathrm{policy}}, \tag{22}$$

$$\underline{C}_{ik,h} = \underline{R}_{ik,h} [\Delta t - t_{ik,h}^{\mathrm{setup}} - t_{ik,h}^{\mathrm{guard}}]_+, \tag{23}$$

$$\sum_j f_{ik,j,h} \le \underline{C}_{ik,h} y_{ik,h}. \tag{24}$$

Use a calibrated payload rate including coding, framing, packet loss and retransmission overhead. For one half-duplex radio,

$$\sum_k y^{\mathrm{TX}}_{ik,h} + \sum_k y^{\mathrm{RX}}_{ki,h} \le 1. \tag{25}$$

Add interference, shared-channel, antenna-pointing and simultaneous-link restrictions as applicable.

For each commodity, maintain a queue with acquisition/processing output, delayed incoming flow and outgoing flow. At a node-time vertex, outgoing bytes cannot exceed bytes available before transmission; propagation and processing delay determine the receiving vertex. Receiver storage and energy constraints apply to every transfer. To accept delivery $d_j = 1$,

$$\sum_{\text{sink arrivals before } d_j^{\mathrm{deliver}}} f_j^{\mathrm{sink}} \ge D_j^{\mathrm{out}} d_j. \tag{26}$$

Bind d_j to an accepted acquisition and valid processing chain. Capturing an image that cannot be delivered must not earn the full delivered-product reward.

> A disconnected physical contact graph cannot be repaired by an optimizer alone. Valid responses are rerouting through real contacts, waiting for a future contact, buffering, compressing, or reducing/deferring acquisition.

7 Complete reference optimization

The mixed-integer decision Π includes observations, processing, supports, contacts, routed flows and resource trajectories. A normalized cost is

$$\begin{aligned}\min_{\Pi} \quad & -\sum_j V_j Q_j^{\mathrm{delivered}} d_j + \lambda_E \sum_{i,h} \frac{E_{i,h}^{\mathrm{consumed}}}{E_i^{\mathrm{usable,max}}} + \lambda_C \sum_{i,h} \frac{C_{i,h}^{\mathrm{executed}}}{C_i^{\mathrm{budget}}} \\ & + \lambda_B \sum_{i,h} \left(\frac{B_{i,h}}{B_i^{\mathrm{safe,max}}} \right)^2 + \lambda_N \sum_{(i,k),h} \frac{\mathrm{airtime}_{ik,h}}{\text{available airtime}_{ik,h}} \\ & + \lambda_A \sum_{g,h} w_g A_{g,h} + \lambda_\Delta \mathrm{ChangeCost}(\Pi, \Pi^{\mathrm{old}}) \\ & + \lambda_R \mathrm{CVaR}_\alpha(L(\Pi, \xi)) + \sum_j M_j s_j.\end{aligned} \tag{27}$$

Only available contacts enter airtime ratios; zero-capacity contacts are excluded. Compute terms and weights use consistent units. Quality is task/mode dependent; products such as Qd are nonlinear unless represented with mode-specific delivery variables.

Constraints include all resource and flow relations above and

$$\sum_{i,h,m} x_{ijhm} \le \kappa_j, \tag{28}$$

$$\sum_{i,h,m} x_{ijhm} + s_j \ge r_j, \qquad 0 \le r_j \le \kappa_j, \tag{29}$$

$$\sum_{j,m,\tau:\ h \in \mathrm{occupy}(i,j,\tau,m)} x_{ij\tau m} \le 1, \tag{30}$$

$$\sum_{j \in O} \sum_{i,h,m} x_{ijhm} \le 1, \quad O \in \mathcal{O}. \tag{31}$$

The conflict family $\mathcal{O}$ handles overlapping task descriptions; intentional redundancy is explicitly exempted. Occupancy includes acquisition, settling and incompatible activity. Use mission slack for unmet demand, not for physical safety.

For persistent monitoring, $A_g(t) = t - t_g^{\text{latest usable acquisition}}$. Receiving an old image resets age to its actual acquisition age, not zero. Define whether usability is local acquisition or ground delivery according to the mission.

The full problem is a mixed-integer nonlinear, uncertainty-aware scheduling/routing problem. It is the reference specification, not the problem to solve globally on every sensor tick.

8 Uncertainty and stale information

For each safety constraint $g_\ell(\Pi, \xi) \le 0$, choose bounded robustness

$$\max_{\xi \in \mathcal{U}_t} g_\ell(\Pi, \xi) \le 0 \tag{32}$$

or a calibrated chance constraint $\Pr\{g_\ell \le 0\} \ge 1 - \epsilon_\ell$. For an approximately Gaussian affine quantity,

$$g_\ell(\Pi, \hat{\xi}) + \Phi^{-1}(1 - \epsilon_\ell)\sqrt{\nabla_\xi g_\ell^\top \Sigma_\xi \nabla_\xi g_\ell} \le 0. \tag{33}$$

This is not automatically valid for heavy-tailed errors, nonlinear failure modes or an uncalibrated covariance. Use conservative energy lower bounds, thermal/storage upper bounds, and contact-capacity lower bounds. Allocate horizon-wide risk using $\sum_{\ell,h} \epsilon_{\ell,h} \le \epsilon_{\text{mission}}$ where the probability model is justified.

For estimate age δ_i, a bounded-drift envelope is

$$\Delta s_i(\delta_i) \le \Delta s_i(0) + L_i \delta_i, \quad E_i^{\text{true}} \ge \hat{E}_i - \bar{P}_i \delta_i - \Delta E_i^{\text{sensor}}. \tag{34}$$

Staleness beyond the certified bound removes a node from new allocations, while its verified safe local behavior may continue.

For S uncertainty scenarios, use

$$\text{CVaR}_\alpha(L) = \min_\eta \left\{ \eta + \frac{1}{(1-\alpha)S} \sum_{s=1}^{S} [L_s - \eta]_+ \right\}. \tag{35}$$

Scenario risk ranks feasible plans; it does not replace hard limits. Avoid claiming out-of-sample guarantees from a small scenario set without additional analysis.

9 Event-triggered repair scope and prefix locking

Maintain an index linking tasks, contacts, buffers, compute stages, nodes and reservations. For event e, compute

$$\mathcal{A}(e) = \text{DependencyClosure}(\mathcal{A}_0(e)), \qquad \mathcal{C}(e) \subseteq \mathcal{V}. \tag{36}$$

A relay failure affects dependent deliveries; a missed downlink affects future acquisitions through storage; a wheel fault affects imaging, charging and antenna access. A fixed two-hop rule is insufficient if dependencies extend farther.

Start with a small patch, expand if necessary, and cap runtime. Freeze unaffected boundary reservations. Coalesce ordinary events using hysteresis, but process safety-critical events immediately. Partition

$$\Pi_i = \Pi_i^{\text{locked}} \oplus \Pi_i^{\text{free}}. \tag{37}$$

Lock executing actions or actions past their practical commitment point. A safety fault may invalidate a locked action: prefix locking prevents unnecessary churn, not emergency cancellation. Revalidate the incumbent after every material event.

10 Distributed allocation using consensus ADMM

10.1 Construct a frozen convex subproblem

Freeze task IDs, topology version and conservative prediction parameters. Precompute windows, use conservative affine approximations where available, relax binary variables to $[0, 1]$, and optimize only the repair patch. Nonconvex pointing and mode decisions must be enumerated, fixed, or safely approximated; they cannot be labelled convex merely by relaxing binary variables.

Let $z \in \mathbb{R}^{Dc}$ hold the patch variables. Each participant holds a copy z_i and solves

$$\min_{\{z_i\}} \sum_{i \in \mathcal{C}} F_i(z_i), \qquad z_i = z_k \quad (i,k) \in \mathcal{E}_{\mathcal{C}}, \quad F_i = f_i + I_{\mathcal{X}_i}. \tag{38}$$

Include each resource cost once. Assign shared constraint rows to deterministic checking nodes holding the necessary variables. Consensus is between copies of the same shared vector, not between different robots' individual assignments. Full copies are limited to the bounded patch, not the entire fleet.

10.2 Updates

For undirected connected coordination edges $e = (i, k)$, use edge variables y_e and scaled duals u_{ie}:

$$z_i^{k+1} = \arg\min_{z_i} \left\{ F_i(z_i) + \frac{\rho}{2} \sum_{e \ni i} \left\| z_i - y_e^k + u_{ie}^k \right\|_2^2 \right\}, \tag{39}$$

$$y_e^{k+1} = \tfrac{1}{2}(z_i^{k+1} + u_{ie}^k + z_k^{k+1} + u_{ke}^k), \tag{40}$$

$$u_{ie}^{k+1} = u_{ie}^k + z_i^{k+1} - y_e^{k+1}. \tag{41}$$

With antisymmetric zero-initialized edge duals, y_e is the endpoint primal average. Warm-start matching variables and compatible reservations; reset incompatible duals after topology or coordinate changes.

In the stacked form $Az + By = 0$, monitor

$$r^{k+1} = Az^{k+1} + By^{k+1}, \qquad s^{k+1} = \rho A^\top B(y^{k+1} - y^k). \tag{42}$$

Use dimension-scaled absolute/relative tolerances and a wall-clock cap. Keep ρ fixed within a batch for a straightforward implementation. An unfinished iterate is a proposal, not an executable schedule.

Classical ADMM results require the relevant closed proper convex functions, feasibility/saddle-point conditions and communication assumptions [6]. They do not prove a global optimum for the integer, nonlinear, evolving mission. Start with reliable versioned rounds during connected windows. On timeout, staleness or topology change, terminate the batch rather than substituting zero for a missing neighbor.

11 Lightweight dual alternative and other monograph methods

For the frozen constraint-coupled relaxation,

$$\min_{x_i \in \mathcal{X}_i} \sum_i f_i(x_i), \qquad \sum_i A_i x_i \le b, \tag{43}$$

choose shares b_i with $\sum_i b_i = b$. Maintain resource prices $\lambda_i \ge 0$:

$$v_i^{k+1} = \sum_{j \in \mathcal{N}_i \cup \{i\}} w_{ij} \lambda_j^k, \tag{44}$$

$$x_i^{k+1} \in \arg\min_{x_i \in \mathcal{X}_i} \{f_i(x_i) + (v_i^{k+1})^\top (A_i x_i - b_i)\}, \tag{45}$$

$$\lambda_i^{k+1} = [v_i^{k+1} + \alpha_k (A_i x_i^{k+1} - b_i)]_+. \tag{46}$$

Use suitable stochastic mixing weights for the stated graph assumptions; arbitrary directed averaging is not interchangeable with doubly stochastic mixing. For applicable frozen convex problems, diminishing steps can use $\alpha_k = \alpha_0/(k+1)^p$, $1/2 < p \le 1$. Primal recovery may need running averages. Finite-budget changing missions require independent feasibility checks regardless of price convergence.

ADMM and the dual method are selectable allocation modes, not simultaneous mandatory layers. Constraint exchange is useful for small shared LPs and active-constraint auditing. Gradient tracking can support a separately formulated smooth cost-coupled calibration problem; its smoothness, step-size and connectivity assumptions do not automatically hold for the main mixed-integer scheduler. Avoid adding algorithms without a specific role.

12 Integer reconstruction and Shen-style bounded local PSO

Independent rounding of fractional assignments can create duplicate tasks or overload resources. Instead, rank candidates by mission priority, relaxed allocation, marginal cost and deterministic tie-breaks; obtain provisional holds; construct a feasible integer schedule; then improve its free suffix.

Most particle evaluations are local, using fixed external reservations and cached resource/contact profiles. A particle is $\pi_p = (a_p, o_p, m_p)$: selected tasks, ordering and modes. Moves include insertion, removal, swap, start shift and mode change. Cross-node reassignment is separately negotiated and requires compatible reservations.

Let $s = k/I_{\max}$:

$$w(k) = w_{\max} - (w_{\max} - w_{\min}) s^\mu, \tag{47}$$

$$c_1(k) = c_{1,\max} - (c_{1,\max} - c_{1,\min}) s, \tag{48}$$

$$c_2(k) = c_{2,\min} + (c_{2,\max} - c_{2,\min}) s. \tag{49}$$

For discrete operator lists,

$$\begin{aligned} v_p^{k+1} = {} & \mathrm{sample}_{w(k)}(v_p^k) \oplus \mathrm{sample}_{c_1(k)}(p_p^{\mathrm{best}} \ominus \pi_p^k) \\ & \oplus \mathrm{sample}_{c_2(k)}(g^{\mathrm{best}} \ominus \pi_p^k). \end{aligned} \tag{50}$$

Define sampling/replication explicitly when a coefficient exceeds one; it is not a probability. This is a proposed decentralized adaptation of Shen's discrete repair structure, not an identical reproduction of their global planner.

After mutation, restore the locked prefix and validate observation windows, slew/activity compatibility, battery through recovery, temperature, processing deadlines, peak storage, routes and reservations. Drop or defer optional tasks if needed; reject any candidate retaining hard violations.

Rank particles lexicographically by (hard violations, unmet critical requirements, objective). Seed with the revalidated incumbent. Preserve it when feasible so search cannot worsen the best-found objective on the same frozen problem. If the incumbent is unsafe, seed with a safe fallback. Terminate on time, iteration or stagnation budget. No global optimum is promised.

13 Discrete glowworm-inspired route repair

Generate at most K_R feasible paths $\mathcal{P}_i^d$ in the time-expanded contact graph. Every candidate respects contact order, capacity, receiver memory, health, energy and deadline. Define

$$C(p) = \alpha_D \widetilde{D}(p) + \alpha_E \widetilde{E}(p) + \alpha_Q \widetilde{Q}(p) + \alpha_F \sum_{e \in p} -\log \underline{p}_e + \alpha_U \widetilde{U}(p) + \alpha_\Delta \mathrm{SwitchCost}(p). \tag{51}$$

The reliability term is a useful score; interpreting it as an exact end-to-end probability requires suitable dependence assumptions. For $J(p) = 1/(1 + C(p))$,

$$L_p^{k+1} = (1 - \rho_L) L_p^k + \gamma_L J(p), \tag{52}$$

$$\mathcal{N}_p^k = \{q : d_{\mathrm{route}}(p, q) < R_p^k,\ L_q^k > L_p^k\}, \tag{53}$$

$$\Pr(p \to q) = \frac{L_q - L_p}{\sum_{r \in \mathcal{N}_p} (L_r - L_p)}, \tag{54}$$

$$R_p^{k+1} = \mathrm{clip}(R_p^k + \beta_R (n^\star - |\mathcal{N}_p^k|), R_{\min}, R_{\max}). \tag{55}$$

Distance is in route-feature space, not arbitrary physical node motion. If the neighborhood is empty, retain the incumbent or use the deterministic best feasible candidate. Return the best validated path found. Enforce TTL/horizon and causal paths; avoid unnecessary repeated physical-node visits.

Switch only if $C(p_{\mathrm{new}}) < C(p_{\mathrm{old}}) - \Delta_{\mathrm{switch}}$, unless the old path fails. This limits flapping. For two or three alternatives, deterministic route selection may be cheaper; keep glowworm search optional and test its incremental value.

14 Task ownership, leases and partition semantics

Numerical consensus is not ownership. Maintain

$$\mathcal{L}_j = (\mathrm{id}_j, \mathrm{owner}, \mathrm{epoch}, t_{\mathrm{start}}, t_{\mathrm{expiry}}, \mathrm{status}, \mathrm{planhash}). \tag{56}$$

Use preconfigured task shards or small replicated committees distributed across agents. A temporary shard leader is not a ground/cloud dependency. Commit membership changes through the selected protocol; disconnected components may not independently redefine the same authority.

The lifecycle is unassigned, held, committed, attempt-consumed, acquired, delivered. Reserve compatible task, contact and receiver-memory resources before execution. Use deterministic acquisition order, expiring provisional holds and durable transaction recovery. An incomplete transaction does not authorize action.

If clock error is bounded by ϵ_t, execution is limited to

$$t_i^{\mathrm{local}} \in [t_{\mathrm{start}} + \epsilon_t, t_{\mathrm{expiry}} - \epsilon_t). \tag{57}$$

If clock uncertainty exceeds the bound, suspend new exclusive acquisitions.

Mode	Uncertain outcome policy	Tradeoff
Strict one-attempt	Persistently consume a unique token before the hardware command; no reissue after uncertain execution.	Avoids repeated authorized attempts but may lose an observation.
Recovery	Retry under timeout/failure policy.	Higher completion; an earlier unreported capture may be duplicated.
Explicit redundancy	Authorize multiple independent captures in advance.	Intentional duplication is budgeted.

A lease prevents overlapping ownership, but cannot prove an expired owner never acquired an image. Strict mode requires durable actuator-side nonces and a consumed-attempt certificate. A crash between consumption and acquisition can lose a task. Recovery mode uses task/content IDs to deduplicate products and measures physical duplicate captures separately.

During a partition, continue safe authorized work and buffer data. Without the required authority, do not issue new globally exclusive grants. Reconcile manifests after reconnection. These properties assume an actuator obeys authorization; a malicious physical controller ignoring commands is outside a software ledger's enforcement guarantee.

15 FDIR and security evidence

For a monitored subsystem, compute residuals and normalized innovation

$$r_{i,k} = z_{i,k} - h_i(\hat{s}_{i,k}), \quad D_{i,k} = r_{i,k}^{\top} S_{i,k}^{-1} r_{i,k}. \tag{58}$$

One persistent detector is $g_{i,k} = \max(0, g_{i,k-1} + D_{i,k} - \nu)$, alarming when $g_{i,k} > \gamma$. Calibrate thresholds against model mismatch, false-alarm costs and the sensor noise distribution.

Event	Immediate response	Repair consequence
Battery reserve threat	Stop optional loads; preserve essentials.	Drop expensive tasks; reconsider charging attitude.
Lost link/radio	Stop failed contact; retain required data.	Repair route and delivery schedule.
Flash pressure	Reject unreserved data; apply retention policy.	Downlink/compress/defer captures.
CPU overrun	Protect control execution.	Change processing mode or defer work.
Overtemperature	Derate or disable affected equipment.	Recompute capabilities and capacities.
Wheel saturation/-fault	Invoke approved platform recovery.	Remove incompatible pointing opportunities.
Navigation uncertainty	Restrict precise actions.	Tighten windows and link predictions.
Suspected compromise	Restrict local interface; gather authenticated evidence.	Constrain suspect routes and claims.

Safe mode is a platform-specific controller with a known operating envelope. The allocator invokes it rather than inventing its control law.

A network anomaly vector may contain packet-rate deviation, authentication failures, replays, sequence gaps, unexpected destinations, queue growth and physical inconsistency. A calibrated detector outputs $\theta_i = P(\text{anomaly} \mid \phi_i)$. Peer evidence can form $\Theta = \sum_i w_i\theta_i / \sum_i w_i$, but this is a score, not Byzantine agreement.

For PBFT-style authorization tolerating f Byzantine members, use $n \geq 3f + 1$ and the actual prepare/commit/view-change rules [7]. Four appropriate members are needed for one Byzantine fault in that model. Do not copy sub-millisecond industrial testbed timeouts to space links. A Byzantine-safe ledger also does not make ADMM Byzantine-resilient: false optimization messages need additional defenses. Local physical validation, rate limits and authenticated state constrain damage, but stronger optimization-level guarantees require separate analysis.

16 Complete algorithm and execution contract

16.1 Persistent local state

Keep state estimates/bounds/timestamps; health and modes; committed plan and fallback; ledger/nonces; contact forecast; route candidates; reservations; data manifest; and compatible warm-start variables. Bound all buffers and logs.

16.2 Fast control and estimation loop

1. Read sensors; validate timestamps, plausibility and authenticity where applicable.
2. Update navigation, attitude, battery, thermal, compute, storage, link and anomaly states.
3. Run the independent safety monitor. Cancel unsafe actions according to platform policy, invoke fallback and emit a critical event.
4. Execute only the next authorized action whose attempt state, reservations and current safety checks remain valid.
5. Update acquisition, processing, transfer, acknowledgement and delivery manifests.

16.3 Planning and repair procedure

1. Validate the incoming event's origin, version, timestamp and priority.
2. Coalesce ordinary events; process safety-critical events immediately.
3. Incrementally update affected predictions and dependency edges.
4. Compute dependency closure and form a bounded repair patch.
5. Preserve unaffected boundary reservations. Separate locked prefix and free suffix. Revalidate the incumbent.
6. Generate physically feasible observation, processing and contact options with uncertainty margins.
7. Construct the frozen convex relaxation with a versioned coordinate map.
8. If a suitable coordination window exists, run ADMM or the selected dual method. Stop on residual tolerance, time cap, topology/version change, stale inputs or timeout.
9. Convert the proposal into an integer allocation through authenticated provisional holds and deterministic contention resolution.
10. Construct a feasible schedule and improve its free suffix using bounded local PSO. Negotiate cross-node changes separately.
11. Repair affected routes through feasible contact-graph alternatives; optionally apply bounded glowworm search.
12. Obtain compatible task, receiver-memory and communication reservations. Abort/release incomplete holds.
13. Independently validate quality/windows, slew/activity compatibility, battery through recovery,

thermal behavior, CPU deadlines, peak RAM/flash, contact capacity/timing and ownership/attempt policy.
14. Commit only a valid versioned plan. Otherwise retain a revalidated incumbent or fallback, expand the patch if budget permits, or explicitly report deferred/missed mission demand.
15. Checkpoint state, expire tentative holds and reconcile manifests after reconnection.

16.4 Required execution invariants

1. No command executes solely because a fractional optimizer variable is positive.
2. Every capture has a valid observation-instance authorization and attempt policy.
3. Every admitted transfer has feasible sender bytes, receiver storage and contact capacity.
4. Essential control has reserved CPU and energy capacity.
5. Current safety validation may override a previously committed action.
6. Every dropped or deferred critical requirement is recorded; physical violations are never hidden by mission slack.

Only the next validated action is executed before replanning or rechecking; the entire horizon is not executed open-loop.

17 Computational, memory and communication budget

Let $c = |\mathcal{C}|$, J be candidate tasks, M operating modes, D relaxed patch dimension and F adjustable schedule length. Allocation communication is approximately $O(K_A|\mathcal{E}_\mathcal{C}|D)$ scalars. Local QP runtime depends on sparsity, conditioning and solver choice; it is not automatically linear in D.

PSO costs $O(PK_PC_{\text{evaluate}}(F))$, often $O(PK_PF^2)$ with straightforward sequence checks. Bound route count and time horizon. Memory includes local/edge optimization vectors, sparse constraints and factorization workspace, particles, sensor ring buffers, manifests and logs. Vector memory alone is not a valid total-RAM estimate.

A 256-element float32 message to four neighbors over eight rounds requires

$$256 \times 4 \times 4 \times 8 = 32768 \text{ bytes} = 32 \text{ KiB} \tag{59}$$

per node per planning epoch, before overhead. At one epoch per second this is about 262 kbit/s; at one epoch per 20 seconds the vector payload alone is about 13.1 kbit/s.

Use sparse windows, bounded patches, changed-coordinate messages where valid, local PSO evaluation, cached predictions, hysteresis and a planning frequency below the state-estimation frequency. Quantization introduces an error floor unless a stronger scheme is analyzed.

Parameter	Initial simulator exploration range
Repair patch	3–8 nodes
Local candidate tasks	8–20
Allocation iterations	5–15
Local PSO particles	8–16
Local PSO iterations	5–15
Candidate routes	2–4
Fine horizon	8–24 slots; separate coarse resource horizon

These are initial experiment settings, not validated flight settings. Derive wall-clock budgets from actual processors and time to the next commitment. For bounded local optimization, the fallback remains available if the budget expires.

18 Worked resource and mesh-healing example

Consider three candidate nodes, safe storage limit 960 MiB and battery reserve 10 Wh. The observation adds 100 MiB of peak storage.

Quantity	A	B	C
Energy after baseline/charging (Wh)	16	24	26
Capture and processing energy (Wh)	9	11	12
Remaining energy (Wh)	7	13	14
Existing storage (MiB)	900	820	880
Peak capture storage (MiB)	1000	920	980
Immediate feasibility	Reject	Accept	Reject

A violates energy and storage. C can become feasible by downlinking 40 MiB first: $880 - 40 + 100 = 940$ MiB. If that downlink costs 2 Wh, its residual is $26 - 2 - 12 = 12$ Wh.

Suppose B's direct ground contact fails. A candidate route is B to C to ground. If B produces 60 MiB, C's receiving occupancy is $880 + 60 = 940$ MiB. B's 2 Wh transmission leaves $24 - 11 - 2 = 11$ Wh. C's receive/forward operation costing 4 Wh leaves $26 - 4 = 22$ Wh. These operations require correctly ordered contacts with enough reserved volume before the delivery deadline.

The planner compares two complete alternatives: B captures while C relays, or C downlinks first and then captures/delivers. Compare delivered utility, delay, total energy and risk. These arithmetic values are illustrative, not simulation results; no optimality or performance improvement is inferred from them.

19 Guarantees, limits and proof obligations

Property	Defensible claim
Resource safety	Conditional on valid bounds, working monitors, reserved resources and a feasible platform fallback.
ADMM convergence	Applies to the frozen convex relaxation satisfying the mathematical and communication assumptions.
Integer feasibility	Established by explicit validation before execution, not by fractional convergence alone.
PSO quality	Bounded-runtime improvement heuristic; no general global optimum.
Mesh healing	Uses alternative real contacts when available within the explored graph/budget.
Ownership	Exclusive authorized ownership under the actual protocol, membership, clock and failure assumptions.
Capture uniqueness	Requires an explicit attempt policy; uncertain retries can create physical duplicates.
Byzantine tolerance	Applies only to the implemented agreement protocol and its fault model, not automatically to optimization.
Mission completion	Impossible to guarantee if energy, observation opportunities, storage or delivery paths are insufficient.

Prefix locking does not prove recursive feasibility. A formal claim needs terminal safe-set or fallback feasibility, bounded disturbances, and a demonstration that the next state remains recoverable. A useful proof decomposition is: local resource-invariance argument; ownership safety argument;

convex-subproblem convergence argument; finite-runtime argument; and separate empirical search-quality assessment. None of these alone proves the full system globally optimal.

20 Simulator validation and experimental plan

Compare independent greedy sensing; resource-aware greedy sensing; distributed allocation alone; allocation plus local PSO; the full design with deterministic route repair; the full design with glowworm search; and a centralized MILP or suitable reference solver on small cases. Hungarian matching is a baseline for simple one-to-one assignment, not an exact solver for the full coupled problem.

Measure delivered utility, completed observations, deadline success, information age, physical duplicate captures, duplicate products, energy per useful delivered observation, minimum reserve margin, overflow, thermal violations, recovery time, optimizer runtime, peak RAM and coordination bytes.

Inject eclipse/harvest forecast error, relay/contact loss, battery degradation, CPU throttling, flash saturation/errors, wheel faults, stale telemetry, partitions, reordered/duplicate messages, failure after capture but before acknowledgement, replay and strategic false reports. Use the same workload/failure traces across methods; report confidence intervals and tail latency, not only averages.

Run ablations disabling each proposed enhancement. Glowworm search is justified only if it improves relevant outcomes over deterministic repair within the same budget. Likewise, compare PSO against simpler insertion/swap search. Check conservation of energy, data and reserved contact volume independently of the scheduler. Persist random seeds and event traces for reproducibility.

21 Recommended implementation sequence

1. Implement truthful physical/resource state transitions, task IDs and data manifests.
2. Add centralized small-instance reference optimization and an independent feasibility validator.
3. Implement distributed allocation with versioned messages and measured communication cost.
4. Add ownership, provisional resource holds, failure/partition semantics and persistent attempt state.
5. Add bounded Shen-style suffix repair and compare it with simple local search.
6. Add contact-graph route repair; introduce glowworm search only as an evaluated alternative.
7. Add uncertainty envelopes, FDIR and adversarial-message experiments.

The strongest core is resource-aware decentralized allocation, bounded local repair, physically valid routing and enforceable ownership. Optional heuristic layers should earn their computational cost through measured benefit.

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